Financial metrics

Every metric on the Lightyear dashboard and in the workbook: the formula, and how to read it.

> **Part 1** is the daily block: the tiles, charts and Core performance table Lightyear builds for every run. **Part 2** is SE Return Analytics, the monthly block. Both are on the Results tab, and the workbook calculates both with live formulas.
>
> There are no run results or examples here. The only numbers are method settings, checked against the code.

Notation used in every formula

Symbol	Meaning
r, b	One day's return on the strategy and on the benchmark, the NIFTY 500 price index.
m	One calendar month's return, compounded from the daily returns. Used in Part 2.
Rf (5%)	The risk-free rate in every Sharpe and Sortino. One config setting. A yardstick, not income.
c	The card's cash rate: what idle cash earns in the backtest. Beta subtracts it. Not the risk-free rate.
V_t	Portfolio value on day t.
mean, stdev	Average and sample standard deviation (divides by n − 1). **Pop stdev** divides by n; Part 2 uses it.
√252, √12	Annualise daily and monthly volatility. 252 is the assumed trading days a year.
years	Calendar days ÷ 365.25. In Part 2, months ÷ 12.

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Part 1. On the Lightyear dashboard (daily)

1.1 The five headline numbers

These tiles appear on every run's Results tab, for the strategy and the benchmark over the same days. The comparators table repeats CAGR, volatility, Sharpe and worst fall for each alternative.

CAGR: the steady yearly growth rate from start to finish

$$\text{CAGR} = \left(\frac{V_{\text{end}}}{V_{\text{start}}}\right)^{\frac{1}{\text{years}}} - 1$$

Volatility: how much returns swing in a year. Lower is a smoother ride

$$\text{Vol} = \text{stdev}\left(r_{\text{daily}}\right) \times \sqrt{252}$$

Sharpe (risk-free rate 5%): return above the risk-free rate, per unit of risk. Higher is better

$$\text{Sharpe} = \frac{\text{CAGR} - 5\%}{\text{Vol}}$$

The 5% is the risk-free rate, a yardstick. It is not income the strategy earns: idle cash earns the card's cash rate. This headline Sharpe is the geometric form. Figures for a single calendar year use the arithmetic form shown in 1.4.

Worst fall from peak (maximum drawdown): the biggest loss from a high to a later low

$$\text{Max drawdown} = \max_{t}\left(1 - \frac{V_t}{\max_{s \le t} V_s}\right)$$

100 became: what 100 grew into, after every daily return

$$\text{Final} = 100 \times \prod_{t}\left(1 + r_t\right)$$

1.2 The charts

Chart	Calculation	How to read it
Growth of 100	100 × running product of (1 + r)	Above the benchmark's line means ahead of it on that day.
Drawdown	V_t ÷ highest V so far − 1	0% is a new high. Deeper dips are bigger losses.
Rolling volatility	stdev of the last 63 daily returns × √252	How bumpy the last three months were.
Calendar-year return	Product of (1 + r) within the year − 1	First and last years may be part-years.
Monthly heat map	Product of (1 + r) within the month − 1	Green months gained, red months lost.

1.3 Core performance table

For the strategy and the benchmark over the same days. CAGR, volatility, Sharpe and maximum drawdown are the formulas in 1.1. The workbook's Core metrics sheet calculates each row with a formula over the Returns

sheet.

Metric	Formula	How to read it
Sortino (risk-free rate 5%)	(CAGR − 5%) ÷ DD, where DD = √mean(min(r − Rf per day, 0)²) × √252	Like the Sharpe, but only falls below the risk-free rate count as risk.
Calmar	CAGR ÷ maximum drawdown	Return per unit of worst fall.
Drawdown duration	Longest run of trading days below a previous high	How long the longest wait to recover took.
95% CVaR (daily loss)	−mean of the daily returns at or below their 5th percentile	The average bad day, as a positive number.
Skew	Sample skew of daily returns	Negative means big falls are more common than big rises.
CAGR without the 10 best days	Remove the 10 best days, then compound; years run from the first return date	Whether a handful of days made the result.
Beta	Cov(r − c, b − c) ÷ Var(b − c)	1.0 moves in line with the index. Below 1 is less sensitive.
Tracking error	stdev(r − b) × √252	How far it strays from the index.
Information ratio	mean(r − b) × 252 ÷ tracking error	Return over the index per unit of straying. Higher is better.
Up / down capture (monthly)	mean monthly return ÷ the benchmark's, in months the benchmark rose / fell	Above 1 up and below 1 down is the good shape.
Rolling one-year periods behind	Share of 252-day windows with compounded return below the index's	How often an investor would have been trailing.

1.4 Figures under the table

Metric	Formula	How to read it
Two-way turnover a year	Σ |weight after − weight before| over every trade, ÷ years	1.0 means trading the whole portfolio's value once a year.
After publication	CAGR, benchmark CAGR, their gap (edge), arithmetic Sharpe and maximum drawdown, on the days from the paper's publication date	Whether the result held once the idea was public. A date given only as a year is read as 31 December.
Sharpe by calendar year (arithmetic)	mean(r − Rf per day) ÷ stdev(r − Rf per day) × √252, on that year's days	The yearly form of the Sharpe. It is labelled arithmetic wherever it appears.
Capacity per Rs 1 crore	For each trade: (its share of capital × Rs 1 crore) ÷ the stock's 20-day average traded value. The 95th percentile P is shown, and the capital at which it reaches the limit: 10% ÷ P	Information only. No fund size is assumed, and the limit is a placeholder.

1.5 Beta in different markets

Metric	Formula	How to read it
Up-market / down-market beta	Beta using only days the index rose / fell	Ideally high in rises and low in falls. The two do not average to the overall beta.

Metric	Formula	How to read it
Beta with the index filter open / shut	Beta using only days the card's index filter was open / shut. The filter's state on a day is read at the previous session's close	Shown when the card has an index filter. A low beta when shut is the filter holding cash.

Part 2. SE Return Analytics (monthly)

A separate block, on monthly returns. It is on the Results tab and in the workbook sheet "SE Return Analytics". Every figure uses calendar-month returns m compounded from the daily returns, the population standard deviation, and three inputs: risk-free rate 5%, Omega threshold 1% a month, drawdown limit 0%. Do not compare a figure here with a daily figure of the same name in Part 1. The benchmark is the price index.

Python and Excel calculate each metric separately. Python calculates it in ros/metrics/se.py; Excel calculates it from a live formula in the workbook. The workbook's "Python vs Excel" sheet sets the two side by side and marks each OK or FAIL at 1e-9.

2.1 Performance

Metric	Formula	How to read it
Total Return	∏(1 + m) − 1	What 1 became over the whole window.
CAGR	$(1 + \text{Total Return})^{12 \div \text{months}} - 1$	Twelve months count as a year, whatever the calendar days.
Avg Month Ret	mean(m)	A typical month.
Positive Months	count of months with m > 0	
Negative Months	count of months with m ≤ 0	A flat month counts as negative.
Max Month Ret	max(m)	
Min Month Ret	min(m)	
Best Month Contri	Max Month Ret ÷ Total Return	How much of the whole gain one month is worth.
Worst Month Contri	Min Month Ret ÷ Total Return	

2.2 Risk Parameters

Metric	Formula	How to read it
Annualized Volatility	pop stdev(m) × √12	Lower is smoother. Population stdev, on months.
Positive Monthly Vol	pop stdev of the months with m ≥ 0	Bumpiness of the winning months. Not annualised.
Negative Monthly Vol	pop stdev of the months with m ≤ 0	Bumpiness of the losing months. Not annualised.
Max Drawdown	min over months of (W ÷ highest W so far − 1), where W = running ∏(1 + m)	A negative number. **The peak starts at the first month's own value, so a loss in the first month is not counted.**
Average Annual Max Drawdown	mean of the Max Calendar Year Drawdown rows	A typical year's worst fall. The same first-month rule applies inside each year, and part-years are included.

Metric	Formula	How to read it
Sharpe Ratio	(CAGR − Rf) ÷ Annualized Volatility	Return above the risk-free rate per unit of risk.
Treynor Ratio	(CAGR − Rf) ÷ Beta	Return above the risk-free rate per unit of market risk.
Sortino Ratio	(CAGR − Rf) ÷ (Negative Monthly Vol × √12)	**Differs from the textbook:** the divisor is the spread of the losing months, not a downside deviation below a target.
Calmar Ratio	CAGR ÷ |Max Drawdown|	Return per unit of worst fall.
Sterling Ratio	CAGR ÷ (|Average Annual Max Drawdown| − drawdown limit)	**Differs from the textbook:** the usual 10-point adjustment is replaced by the drawdown limit, which is 0.
Omega Ratio	Σ(m − T) over months with m > T ÷ |Σ(m − T) over months with m < T|, T the monthly threshold	Above 1 means gains over the threshold outweigh shortfalls.
Gain to Pain ratio	Σ of months with m ≥ 0 ÷ |Σ of months with m < 0|	Winning months against losing months.
Tail Ratio	|95th percentile of m| ÷ |5th percentile of m|	Above 1 means big wins are larger than big losses.

2.3 Performance Vs Index

Metric	Formula	How to read it
Beta	slope of m on the benchmark's monthly returns	1.0 moves in line with the index.
Alpha	CAGR − $CAGR_{benchmark}$	Points a year ahead of the index.
Jensens Alpha (Adj Beta)	CAGR − $CAGR_{benchmark}$ × Beta	**Differs from its label:** no risk-free rate and no adjusted beta are used.
Positive Monthly Alpha	count of months with m > b	Months ahead of the index.
Negative Monthly Alpha	count of months with m < b	Months behind it.
Tracking Error	pop stdev(m − b) × √12	How far it strays from the index.
Information Ratio	Alpha ÷ Tracking Error	The CAGR gap, not the mean monthly gap, over tracking error.
Hit Rate	Positive Monthly Alpha ÷ (Positive + Negative Monthly Alpha)	Share of months ahead; ties are left out.
Max Monthly Alpha	max(m − b)	
Min Monthly Alpha	min(m − b)	

2.4 Capture Ratio

Metric	Formula	How to read it
Ret +Ve	mean(m) in months with b ≥ 0	What it earned in the index's up months.
Ret -Ve	mean(m) in months with b < 0	What it lost in the index's down months.

Metric	Formula	How to read it
Upside Capture	Ret +Ve ÷ the benchmark's Ret +Ve	Above 1 rises more than the index.
Downside Capture	Ret −Ve ÷ the benchmark's Ret −Ve	Below 1 falls less than the index.
Capture Ratio	Upside Capture ÷ Downside Capture	Above 1 gains more than it gives back.

2.5 Extreme Capture Ratio

Metric	Formula	How to read it
Ex. Ret +Ve	mean(m) in months with b ≥ 5%	The index's strongest months.
Ex. Ret -Ve	mean(m) in months with b < −5%	The index's worst months.
Ex. Upside Capture	Ex. Ret +Ve ÷ the benchmark's	
Ex. Downside Capture	Ex. Ret −Ve ÷ the benchmark's	
Extreme Capture Ratio	Ex. Upside Capture ÷ Ex. Downside Capture	Whether it holds up when the market moves hardest.

2.6 Tail Risk

Metric	Formula	How to read it
Skewness	sample skew of m	Negative means big falls are more common than big rises.
Kurtosis	sample excess kurtosis of m	A bell curve scores 0. Positive means fat tails.
95% VaR	5th percentile of m	A negative return, not a positive loss.
99% VaR	1st percentile of m	A negative return.
95% CVaR	mean of the months with m ≤ 95% VaR	The average month once you are in that tail.
99% CVaR	mean of the months with m ≤ 99% VaR	

2.7 Period tables

Metric	Formula	How to read it
Quarterly return	∏(1 + m) over the calendar quarter − 1	A quarter with fewer than three months is a part-period.
Financial Year Performance	∏(1 + m) from April to March − 1	* marks the first year, ** a last part-year.
Calendar Year Performance	∏(1 + m) from January to December − 1	Same marks.
Max Calendar Year Drawdown	The Max Drawdown formula on that year's months alone	**A loss in the year's first month is not counted**, because the peak starts at that month's own value.

Sources: ros/metrics/core.py, ros/metrics/se.py, lightyear/charts.py, lightyear/workbook.py, lightyear/se_sheet.py, config/fund.yaml, config/gate_thresholds.yaml.